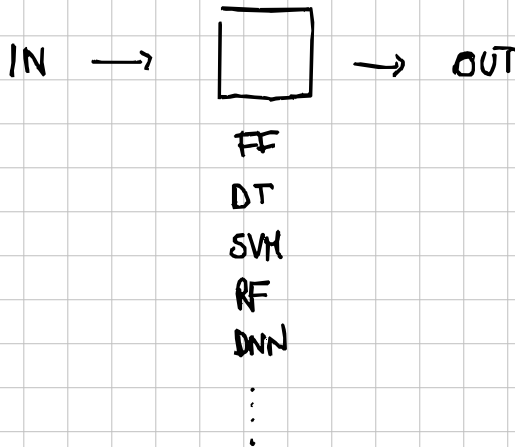
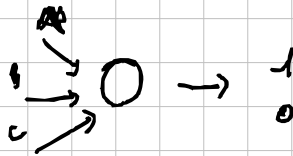


Statistical Physics of Learning (Michael Biehl)

A bit of history...



▲ McCulloch & Pitts (1943) : neuron



▲ Hobb : synaptic plasticity (1949)

"Five together, wire together"



▲ John Hopfield → NN
(Amari, Little)

AUTO-ASSOCIATIVE
MEMORY

Energy driven dynamics, analogy
with magnet materials, (spin-glass)

The space of interactions in NN (Gordner)

UNIVERSALLY VALID PRINCIPLE / ERROR BOUNDS



STATISTICAL PHYSICS
OF LEARNING

OBSERVATIONS / CONCRETE APPLICATIONS

Ex. Vapnik - Chervonenkis theory

[books: - Introduction to The Theory of Neural Computation
- Statistical Mechanics of Learning (Engels)
- Information, Physics and Computation]

The Perceptron (1957) [Rosenblatt]

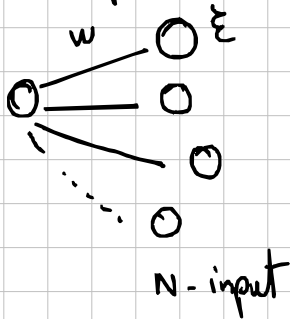
[Perceptrons: An introduction to computational geometry]

Adaptive Linear Neuron - Adeline

Basic concepts of SPL

- Linear separability
 - Rosenblatt's perceptron algorithm
 - Learning in version space
- Student / Teacher scenario
 - HD data, large Networks, generalization error
- typical learning curves
 - ONLINE / OFFLINE LEARNING

The Perceptron



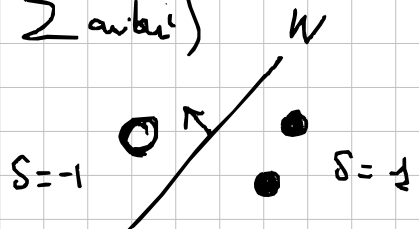
$$S = \text{sign}(W \cdot x - \theta) = \pm 1$$

- Perceptron Convergence Theorem [Rosenblatt]
- Capacity [Cover Theorem]

Linearly separable Function

$$S(x) = \text{sign}(w \cdot x) \quad \forall w \in \mathbb{R}^n$$

$$(a \cdot b = a^T b = \sum a_i b_i)$$



$$D = \{ \mathbf{z}^p, S^p = \text{sign}(w \cdot \mathbf{z}^p) \}_{p=1}^P$$

↳ linearly separable algorithm

Given D , initialize $w(0) = \emptyset$,

present $\{ \mathbf{z}^t, S^t \}$ in order

$$t = \underbrace{1, \dots, P}_{\text{epoch}}, 1, \dots, P, 1, \dots, P$$

$$w(t) = \begin{cases} w(t-1) & \text{CORRECT} \\ w(t-1) + \frac{1}{N} \mathbf{z}^t S^t & \text{Else} \end{cases}$$

Hebbian $\rightarrow$ INPUT $\times$ OUTPUT
(Fire together, wire together)

Perceptron Convergence Theorem

IF $D = \{ \mathbf{z}^p, S^p \}$ is linearly separable

Rosenblatt's perceptron converges after a finite number of epochs with $\text{sign}(w \cdot \mathbf{z}^p) = S^p$

$\forall p = 1, \dots, P.$

↳ How long will it take?

Consider a rule $S_R(z) = \text{sign}(w^* \cdot z)$
 defined by a specific vector w^* TEACHER

Consider a Training set $D = \{z^p\}_{p=1}^P$,

Find a $\bar{w}$ (student) with a consistent hypothesis S_H with

$$S_H(z^p) = S_H^p = \text{sign}(w \cdot z^p) = S_R^p \quad 1 \dots P$$

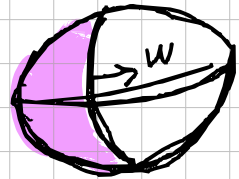
→ Learning a **VERSION SPACE**

$$V_D = \{w \mid \text{sign}(w \cdot z^p) = S_R^p\}$$

Each pair $\{z^p, S^p\}$ defines an HYPERPLANE
 in $\mathbb{R}^N$ which separates correct from incorrect

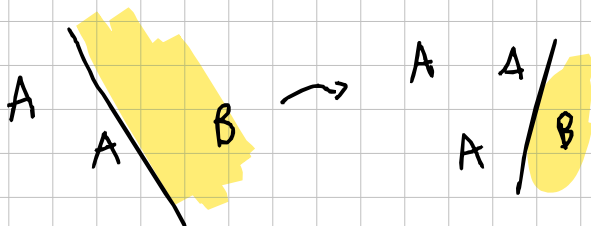
$$\text{sign}(w \cdot z^p) = S$$

$$\text{sign}(w \cdot z^p) = -S$$



TEACHER "somewhere" in **VERSION SPACE**

→ Smaller Version space



So W student $\rightarrow W^*$ Teacher

wants a degree of agreement between W & W^*

- Consider random Test input, draw $P(z)$

$$E_g = \text{Prob}(S_H(z) \neq S_R(z))$$

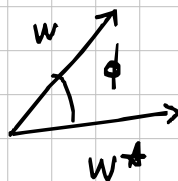
- $P(z)$ spherically symmetric $\langle z_i \rangle = 0$

GENERALIZATION ERROR

$$\langle z_i z_k \rangle = \delta_{ik}$$

$$- E_g = \frac{2\phi}{2\pi} = \frac{1}{\pi} \arccos \left[\frac{W \cdot W^*}{\|W\| \|W^*\|} \right]$$

$$= \begin{cases} 0 & \text{perfect } W \parallel W^* \\ 1/2 & W \perp W^* \end{cases}$$



CENTRAL LIMIT THEOREM

Consider iid random numbers x_i $\langle x_i \rangle = \mu$

$$Y = \sum x_i \sim N(\mu, \sigma^2) \quad n \rightarrow \infty \quad \langle x_i \rangle = \mu, \quad \langle x_i^2 \rangle = \sigma^2$$

independently of the details of x_i

(can be extended of course)

Thermodynamic Limit

HD spaces $N \rightarrow \infty$

$$w, w^*, z \in \mathbb{R}^N$$

$$|w|^2 = o(N)$$

$$|w^*|^2 = o(N)$$

Let's draw input vectors $\langle z_j \rangle = 0$

$$z^2 = \sum z_j^2 = N$$

$$\langle z_i z_j \rangle = \delta_{jk} = \begin{cases} 1 & j=k \\ 0 & j \neq k \end{cases}$$

↓
UNCORRELATED

Ex. $P(z) = N(0, 1) \Rightarrow z_j = \begin{cases} 1 & P=1/2 \\ -1 & P=1/2 \end{cases}$

$$h = \frac{w \cdot z}{\sqrt{N}} = \frac{1}{\sqrt{N}} \sum_i w_i z_i$$

joint normal density

$$b = \frac{w^* \cdot z}{\sqrt{N}} = \frac{1}{\sqrt{N}} \sum_i w_i^* z_i$$

$$\langle b \rangle = 0 = \frac{1}{\sqrt{N}} \sum w_i \langle z_i \rangle = 0$$

$$\langle h \rangle = 0$$

ORDER PARAMETERS

ASSUMED
FIXED

$$\langle b^2 \rangle = \langle z_i z_k \rangle = \frac{1}{N} \sum (w_i^*)^2 = T = 1$$

$$\langle h^2 \rangle =$$

$$= \frac{1}{N} \sum w_i^2 = Q$$

$$\langle hb \rangle =$$

$$= \frac{1}{N} \sum w_i w_i^* = R$$

$$P(h, b) = \frac{1}{2\pi \sqrt{\det C}} \exp \left[-\frac{1}{2} \begin{pmatrix} h - \langle h \rangle \\ b - \langle b \rangle \end{pmatrix}^T C^{-1} \begin{pmatrix} h \\ b \end{pmatrix} \right]$$

$$C = \begin{pmatrix} Q & R \\ R & 1 \end{pmatrix} \rightarrow \text{macroscopic parameters}$$

$$= \frac{1}{2\pi \sqrt{Q - R^2}} \exp \left[-\frac{1}{2} \frac{h^2 + Qb^2 - 2Rhb}{Q - R^2} \right]$$

$$\begin{aligned} E_g = \text{Prob}(hb < 0) &= \langle \Theta[-hb] \rangle = \int_0^\infty dh \int_{-\infty}^0 db \\ &\quad + \int_0^\infty db \int_0^\infty dh P(h, b) \\ &= \frac{1}{\pi} \arccos \left[R/\sqrt{Q} \right] = \frac{1}{\pi} \arccos \left[\frac{w \cdot w^*}{\|w\| \|w^*\|} \right] \end{aligned}$$

We find the same result in HD

↳ HD space described with same order parameter

Remarks with learning in a version space

- Linear separability of the rule ??
- " " of the data ??
- Is data reliable ??

We should

- Tolerate errors
- aim at small # of errors
- apply heuristic algorithms
- SVM : nonlinear

19/08/2026

Again $\frac{1}{N} \sum_{j=1}^N w_j^2 = Q$

$$\frac{1}{N} \sum_{j=1}^N w_j w_j^* = R$$

(1) Characteristic properties with many (MICRO) degrees of freedom fully described by few parameters (MACROSCOPIC)

Online Perceptron Training

- UNCORRELATED EXAMPLES
- NO REPETITIONS

Training set $D = \{z^p, s_R^p\}_{p=1}^P$ of reliable examples with a teacher-student setup $s_R(z) = \text{sign}(w^* z)$

- $w(0) = 0$ $t = p$ ONLINE
 - present $\{z^p, s_R^p\}$ at a time, no repetition → step size
 - update with $w(p) = w(p-1) + \frac{\eta}{\sqrt{N}} f(w(p-1))$
- $z^p s_R^p$

→ Hebbian Learning

η = learning rate
 $f(w(p-1))$ modulating function

→ Stop after P examples

$$w^2 \propto O(N)$$

What is F modulating function?

$$F \propto Q^{p-1} = w(p-1)^2 / N$$

$$h^p = w(N-1) \cdot z^p / \sqrt{N}$$

$$S_R^p = \text{sign}(b^p)$$

AVAILABLE
INFO

$$f(Q^{p-1}, h^p, S_R^p)$$

Microscopic

$$(A) \quad w(p) = w(p-1) + \frac{\eta}{\sqrt{N}} f(Q^{p-1}, h^p, S_R^p) z^p S_R^p$$

$$\rightarrow \frac{w}{\sqrt{N}} \text{ on (A)}$$

$$\rightarrow \underbrace{\frac{w(p) \cdot w^*}{N}}_{R(p)} = \underbrace{\frac{w(p-1) w(p)}{N}}_{R(p-1)} + \frac{\eta}{N} f(\dots) \underbrace{\frac{w^* z^p}{\sqrt{N}} S_R^p}_{b^p}$$

→ Square

$$\rightarrow Q(p) = Q(p-1) + \frac{2\eta}{N} f(\dots) h^p + \frac{\eta^2}{N^2} f^2(\dots)$$

Recursion relation For order parameters

$$R(p) = R(p-1) + \frac{\eta}{N} f(\dots) b^p \text{sign}(b^p)$$

$$Q(p) = Q(p-1) + \frac{2\eta}{N} f(\dots) b^p \text{sign}(b^p) + \frac{\eta^2}{N} f^2(\dots)$$

Average $\langle \rangle_p$ in terms of Gaussian integrals

$$\langle R(p) \rangle_p = R(p-1) + \frac{\eta}{N} f(\dots) \langle b^p \rangle \text{sign}(b^p)$$

General Form

$$R(p) = R(p-1) + \frac{1}{N} F_R [R(p-1), Q(p-1)]$$

$$Q(p) = Q(p-1) + \frac{1}{N} F_Q [R(p-1), Q(p-1)]$$

Only local step, but over all history?

(2) Self averaging of order parameters
For $N \rightarrow \infty$

$$\begin{aligned} \rightarrow \text{Consequences } & [F_{R,Q}(R(p-1), Q(p-1))] \\ & = F_{R,Q}([R(p-1)]_{p-1}, [Q(p-1)]_{p-1}) \end{aligned}$$

$$R(p) = R(p-1) + \frac{1}{N} F_R(R(p-1), Q(p-1))$$

$$Q(p) = \dots$$

Continuous time limit $\alpha = \frac{\mu}{N}$

$$\mu \rightarrow \mu + 1$$

$$\alpha \rightarrow \alpha \rightarrow \frac{\alpha + 1}{N}$$

$$\frac{1}{N} = d\hat{\alpha}$$

$$\alpha \rightarrow \alpha + d\hat{\alpha}$$

(4) ODE

$$\frac{R(\mu) - R(\mu-1)}{1/N} = F_R(R(\mu-1), Q(\mu-1))$$

$N \rightarrow \infty$

$$\frac{dR}{d\alpha} = F_R(R, Q)$$

$$\rho = \frac{R}{\sqrt{Q}}$$

$$\frac{d\rho}{d\alpha} = \frac{F_R(R, Q)}{\sqrt{Q}} - \frac{1}{2Q} \rho F_Q(R, Q)$$

Define $R(0), Q(0), \rho(0)$
 $R(\alpha), Q(\alpha), \rho(\alpha)$ numerically
integrate them

learning curves

Rosenblatt's perceptron algorithm

$$f(\dots) = \Theta(h^\mu b^\mu) = \begin{cases} 1 & \text{if } \text{sign}(h^\mu) \neq \text{sign}(b^\mu) \\ 0 & \text{else} \end{cases}$$

$$\eta = 1$$

$$h, b \sim \mathcal{N}$$

$$\langle \Theta(-hb)^2 \rangle = \frac{1}{\pi} \arccos\left(\frac{R}{\sqrt{Q}}\right)$$

$$\begin{cases} \frac{dp}{d\alpha} = \frac{1}{\sqrt{2\pi}} \left(\frac{1-p^2}{\sqrt{Q}} - \frac{p}{Q} \arccos(p) \right) \\ \frac{dQ}{d\alpha} = \sqrt{\frac{2}{\pi}} \left((p-1)\sqrt{Q} + \frac{1}{\pi} \arccos(p) \right) \end{cases}$$

$$\text{As } \alpha \rightarrow \infty, R, Q \rightarrow \infty \quad w \xrightarrow{||} w^*$$

$$\lim_{\alpha \rightarrow \infty} p(\alpha) = 1 \quad E_g(\alpha) \sim \frac{1}{\pi} \left(\frac{2}{3}\right)^{-1/3} \alpha^{-1/3} \quad \text{as } \alpha \rightarrow \infty$$

Adaptive Reception

→ Learning from mistakes $w \cdot \sum^n \cdot S_R^n < 0$

→ Adaptive update $|w \cdot \sum^n|$

Semi-Linear modulation
Function

$$f(\dots) = (-h^p \text{sign}(b^p))$$
$$\approx [-h^p \text{sign}(b^p)]$$

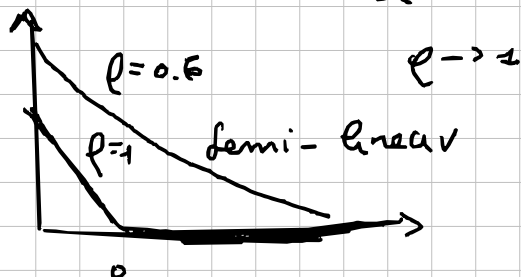
$$w(p) = w(p-1) - \frac{1}{\sqrt{N}} (h^p S_R^p) \sum^n S_R^n$$

$$\dots \quad E g(\alpha) \approx \frac{3}{2\alpha}$$

Maximize $\frac{dp}{d\alpha} \quad ?? \quad F_{opt} \propto Q, R$

- Requires knowledge of Q & R
- ρ as a parameter

Analysis gives us $E g(\alpha) \approx \frac{0.88}{\alpha} \quad \alpha \rightarrow \infty$



- Learning From queries : students select $z^N \perp w(p-1)$, "students have more doubts", there is

$$E_g = \frac{1}{\pi} \exp(-x/\pi)$$

- Learning From noisy data : stochastic corruption in the training data

- Input noise
- Output noise

- Learning drifting concepts : teacher weights change during training

- Learning From structured data (non isotropic)

OFFLINE LEARNING choice of w guided by a fixed batch of examples

TRAINING SET $D = \{z^p, S^p, R^p = SR(z^p)\}_{p=1}^P$

aim at a minimization of a Energy Function

$$E(w, Z, SR) = \bigoplus_{p=1}^P \left(-(w \cdot z^p)_{SR} \right) = \begin{cases} 0 & \text{sign}(w \cdot z) = SR \\ 1 & \text{otherwise} \end{cases}$$

COST FUNCTION / TRAINING : $H(w) = \sum_{p=1}^P E(w, z^p, SR^p)$
 number of misclassified elements

$H(w)$ with $w \in \mathbb{R}^N$ $w \in \{-1, +1\}^N$

→ STAT PHYS OF STOCHASTIC OPTIMIZATION
METROPOLIS - HASTINGS ALGORITHM

$w \rightarrow \tilde{w}$
 $\tilde{w}_j \rightarrow -w_j$ → compute $\Delta H = \hat{H}(w) - H(w)$

$\Delta H < 0$ always

$\sim \exp(-\beta \Delta H)$

→ CONTROL
PARAMETER

Langevin Dynamics

$$\frac{\partial w}{\partial t} = -\nabla_w H(w) + F(t)$$

→ noise (white)

$$\langle f_i, g_i \rangle = \delta_{ij} \delta(t-s)$$

→ How much deviation from gradient descent

Thermal equilibrium Gibbs-Boltzmann

$$P_{eq}(w) = \frac{1}{Z} \exp[-\beta H(w)] \quad Z = \int dw \exp[-\beta H(w)]$$

$$[dw(w) = d^N w \delta(w^2 - N)] \quad \rightarrow \text{STATIONARY density STATES}$$

$$\langle H \rangle_T = \frac{1}{Z} \int dw(w) H(w) \exp[-\beta H(w)] = \frac{-\partial \ln Z}{\partial \beta}$$

$$Z = \int dE \int dp(w) \delta[H(w) - E] e^{-\beta E}$$

$$\equiv e^{Ns(E)} \sim \text{states with } H = E$$

$$s = \frac{1}{N} \int dp(w) \delta[H(w) - E] \quad \text{MICROCANONICAL entropy}$$

Assume EXTENSIVE $E = N$:

$$Z = \int dE \exp[-N\beta (e - s(e)/\beta)]$$

$N \rightarrow \infty$ P is dominated by states with minimal energy

$$f = e - s/\beta = -\ln Z / (\beta N)$$

↓

Free energy density

How many states exist ?? - low energy
- available state

In the limit $T \rightarrow \infty$ (temp)

$f \approx e$ grandstates

$T \rightarrow \infty$

Languin $\rightarrow$ Real dynamics	Noise dominates Random changes
Metropolis $\rightarrow$ Only downhill	

Adaptive $w \in \mathbb{R}^N$ guided by

$$H(w) = \sum E(w, \mathcal{Z}^p, S_R^p) \text{ given a}$$

specific Dataset $P = \alpha N \rightarrow$ extensive energy

• $\bar{w}$ comes out of an optimiz. process

• In equilibrium with $\ln Z$

• Thermal averages, For a specific dataset

We want $\langle \ln Z \rangle_D$ ^{QUENCHED} Free energy
 _{$\frac{1}{N}$} $\hookrightarrow$ all datasets

$\rightarrow$ Training at $\beta \rightarrow 0$: weak influence of individual data
but MANY examples $P = \alpha N$ $\alpha \rightarrow \infty$
 $\tilde{\alpha} = (\beta \alpha) \approx 1$

There it's true that $\langle \ln Z \rangle_D = \ln \langle Z \rangle_D$
(annealed approx)

We sample input very well

$$\langle H(w) \rangle_T \rangle_D = P \langle E(w, \mathcal{Z}, S_R) \rangle_{\mathcal{Z}, S_R}$$

$$\frac{\ln \langle Z \rangle_D}{N} = F \rightarrow \text{FREE energy IN HWT}$$
$$\beta F = \beta \alpha E_g - S(E_g)$$

Ex. Perceptron Learning

$$S = \text{sign}[w \cdot \vec{z}]$$

$$Q = 1$$

$$T = \text{sign}[w^\star \cdot \vec{z}]$$

$$w^\star \cdot w = N \quad T = 1$$

$$D = \{ \vec{z}^p, S^p \}_{p=1}^P$$

$$\langle \vec{z}_j \rangle = 0$$

$$\langle \vec{z}_i \vec{z}_j \rangle = \delta_{ij}$$

$$h = \frac{w \cdot \vec{z}}{\sqrt{N}}$$

$$b = \frac{w^\star \cdot \vec{z}}{\sqrt{N}}$$

$$R = \frac{w \cdot w^\star}{N} = \rho \sim O(1)$$

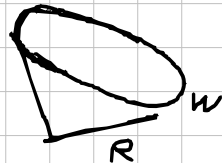
$$E_g = \frac{1}{\pi} \arccos[R]$$

Entropy

weights with order parameter R

$$r \sim \sqrt{1 - R^2}$$

$$V \sim (1 - R^2)^{N/2}$$



$$S(R) = \frac{1}{N} \ln \int \text{Volume} = \frac{1}{2} \ln [1 - R^2] + C$$

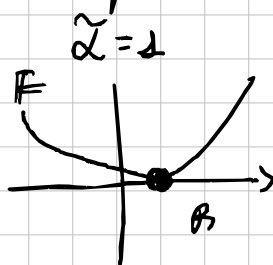
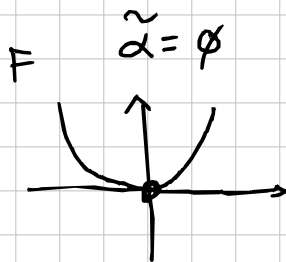
General

$$C = \begin{bmatrix} 1 & R \\ R & 1 \end{bmatrix} \Rightarrow S(R) = \frac{1}{2} \ln \det(C)$$

$$\text{HT Free energy} \quad \beta F = \tilde{\alpha} E_g(R) - S(R)$$

physical state is minimum of βF

$$\frac{\partial F}{\partial R} = 0$$



Limits;

- T and N are not independent
- Training error and generalization error are not separable

DISORDERED AVERAGE: Replica trick

$$\begin{aligned} n \in \mathbb{R} \quad Z^n &= \exp[n \ln Z] \approx 1 + n \ln Z \\ &= \langle \ln Z \rangle_D = \lim_{n \rightarrow 0} = \lim_{n \rightarrow 0} \frac{\partial \langle Z^n \rangle_D}{\partial n} \end{aligned}$$

What is Z^n ?

$$Z^n = \int \left(\prod_{j=1}^n d\mu(\omega^j) \right) \exp \left[-\beta \sum_{j=1}^n H(\omega^j) \right] \quad \begin{matrix} N \text{ copies} \\ \text{NOT INTERACT} \end{matrix}$$

$$\begin{aligned} \langle Z^n \rangle &= \left\langle \exp \left[-\beta \sum_{j=1}^n H(\omega^j) \right] \right\rangle \\ &\approx 1 - \beta \sum_{j=1}^n \langle H(\omega^j) \rangle_D \quad \beta \rightarrow 0 \end{aligned}$$

REPLICAS DECOUPLE HIGH T

if $\beta > 0$, coupling replicas
 $= \exp[-\beta \text{Heff}(w^1, w^2, \dots, w^n)]$

$$\langle z^n \rangle_D \quad q_{st} = \frac{w^s \cdot w^t}{N} \quad R_s = \frac{w^s \cdot w^*}{N}$$

$s, t \in \{1, 2, \dots, n\}$

■ Replica symmetry

Zero learning with $\beta \rightarrow \infty$